USER GUIDE

Poseidon

3D Asset Toolkit & Doctor

Poseidon is a desktop toolkit that helps 3D artists, technical directors and game developers **clean up, check and package assets** before they go into Unreal Engine 5 or Unity: batch renaming, fixing texture paths inside FBX, consolidating .blend textures, previewing meshes and textures, ORM packing, texture conversion, finding duplicate or unused files, matching meshes to textures and assembling a clean kit.

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The same content as the in-app Help (press F1).

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1 Getting started

What Poseidon is, the layout, drag & drop, shortcuts, the command palette, the log and appearance.

What is Poseidon?

Poseidon is a desktop toolkit that helps 3D artists, technical directors and game developers **clean up, check and package assets** before they go into Unreal Engine 5 or Unity: batch renaming, fixing texture paths inside FBX, consolidating .blend textures, previewing meshes and textures, ORM packing, texture conversion, finding duplicate or unused files, matching meshes to textures and assembling a clean kit.

- Most writing operations have a preview step (**Preview** / **Dry run**), a `.fbx.bak` backup or **Undo**. The duplicate / unused finders only report; they never delete.
- The app's interface is in English; this guide shows button and option names in **bold** exactly as on screen.

Layout

- **Navigation rail (left)**: the 8 tools, in the order of Ctrl+1 … Ctrl+8. At the bottom: **Appearance** (gear), **About & contact**, **Command palette** and the log toggle.
- **Header**: the current tool, the **Search actions…** box (opens the command palette), the **Assets** and **Broken paths** counters (broken texture paths found by FBX Doctor), the **?** button for this guide, and the version pill (click it to check for updates).
- **Log (bottom)**: activity from the app and the Rust backend, including Blender's output.

Opening files and folders

- **Drag & drop** files or folders anywhere on the window: they load into the tool that is open.
- Exceptions: dropping a **single** `.blend` file switches to Blender Packer; dropping one mesh / texture while in Batch Renamer (or a non-FBX file while in FBX Doctor) opens it in Asset Previewer.
- Each tab has its own open buttons (**Open folder**, **Files**, **Choose…**…). Ctrl+K → **Open folder…** / **Open files…** also sends the result to the current tab.
- Dashboard and Asset Matcher have **Choose…** plus a ☁ menu: Google Drive for desktop drives (**My Drive**, **Shared drives**) or a pasted folder path. A Drive web link can't be opened directly; the menu explains how to reach it through Drive for desktop. Listing folders downloads nothing.

Keyboard shortcuts

`F1`	Open / close this guide (on the current tab's topic)
`Ctrl` `K` / `Ctrl` `P`	Command palette
`Ctrl` `1` … `Ctrl` `8`	Switch tool, in navigation-rail order
`Ctrl` `` ` ``	Show / hide the log
`Ctrl` `,`	Appearance: theme, fonts, interface size

`Esc`	Close a dialog / the command palette

Command palette

Press Ctrl+K (or click **Search actions…**), type to filter, ↑ / ↓ to choose, Enter to run, Esc to close.

- Commands: **Go to …** (each tool), **Open folder…**, **Open files…**, **Toggle log drawer**, **Assemble kit from manifest…**, **Appearance settings…**, **Toggle light / dark theme**, **About & contact**, **Check for updates…**, **Clear log** and **Help / Hướng dẫn**.

Log

- Click the level badges (debug, info, warn, error) to hide / show a level; the trash icon clears the log.
- Drag the log's top edge to resize it. It follows new lines unless you scrolled up to read history.
- When something fails, check the log first: most tools write the details there.

Appearance and updates

- **Appearance** (Ctrl+, or the gear): **Dark** / **Light** / **System** theme, **Interface font**, **Code & path font**, **Interface size** 90–125 %. **Reset** restores the defaults. Settings are remembered.
- The installed app looks for a new GitHub release a few seconds after start; check manually with the header's version pill or Ctrl+K → **Check for updates…**.
- The guide's language (VI / EN, top right) is remembered too.

2 Batch Renamer

Batch-rename meshes and textures to UE5, Unity or custom rules, with a live preview of every new name and undo.

How to use

1. Drop a folder (or some files) onto the tab, or click **Open folder** / **Files**. One folder at a time.
2. Choose the file type: **All**, **Meshes** (FBX, OBJ, glTF, GLB), **Textures** or **PSD**. Tick **Subfolders** to include sub-folders (not available for individual files).
3. In **Rules**, click **Add** to add a rule. An Unreal Engine 5 preset is there by default.
4. The **Preview** table updates as you edit rules: **Current name** → **New name**, status and note.
5. Click **Rename N files** to rename the OK rows. **Undo** reverts the latest rename (the number in brackets is how many can be undone).

Rule types

- **Preset: Unreal Engine 5** — meshes become `SM_Name` (or `SK_Name` when the name hints at a skeletal mesh: skel, skeletal, skinned, sk), textures become `T_Name_D` / `_N` / `_R` / `_M` / `_ORM`.
- **Preset: Unity** — PascalCase, textures `_Albedo` / `_Normal` / `_MaskMap`.
- **Search & replace** — **Find** / **Replace**; tick **Regex** for regular expressions (groups `$1` or `${1}`); **Match case** makes it case-sensitive.
- **Prefix** / **Suffix** — add text before / after; **Skip names that already have it** leaves those names alone.
- **Change case** — snake_case, camelCase, PascalCase, UPPER_CASE, lowercase, kebab-case.
- Rules run top to bottom on the file name without its extension. The checkbox on a card enables / disables it; arrows reorder; the trash icon removes it.

Preview statuses

- **OK** — will be renamed. **Skipped** (counted as unchanged) — the name already matches.
- **Collision** — the new name clashes with another file. **Invalid** — the new name is not valid; see the **Note** column.
- **Hide unchanged** (on by default) hides unchanged rows. The table shows the first 1,500 rows, but every file is renamed.

Safety

- Nothing is written until you click **Rename**. Collisions are checked up front and again right before renaming (refused files are reported in the log).
- Renames go through temporary names in two phases, so swaps like a → b, b → a are safe.
- History is kept in `.rename_history.json` in the working folder, so **Undo** still works after restarting the app. The Recipes **Rename assets** step is undone from this tab too.

3 FBX Doctor

Find and fix absolute or broken texture paths inside FBX files (binary 7.x and ASCII) without opening a 3D app.

How to use

1. Drop FBX files or folders (searched recursively) onto the tab, or click **Open folder** / **Open FBX**.
2. The **Files** list: green = portable, amber = has issues (the red number counts bad paths), red = unreadable. The header's **Broken paths** counter is the total.
3. Select a file to see its table: **Object / field**, **Path in file**, **Kind** (Absolute, UNC, Relative, Empty) and **On disk** (Found, By name, Missing, Embedded).
4. In the **Fix** panel, choose the **Mode**, options and scope **All files** / **Selected file**.
5. Click **Preview** to see every change (old path struck through → new path), then **Fix N files** to write.

Fix modes

- **Relative** — a path from the FBX to where the texture really is. **Rebase to folder (optional)** points at a fixed folder instead of each texture's found location.
- **Strip** — keep only the file name (e.g. `wood.png`).
- **Collect** — copy the textures into a sub-folder next to the FBX (**Texture subfolder**, default `textures`) and point there.

Options

- **Extra texture folders** → **Add folder**: textures are looked up by file name in these folders (up to 4 levels deep), e.g. a sibling Textures folder. Matches found this way show as **By name**.
- **Also search the parent folder** (on by default): also finds `../Textures` next to the FBX folder.
- Scope **All files** = every file with issues (every readable file when **Rewrite every path** is on); **Selected file** = only the selected one.
- **Rewrite every path** — also rewrites valid paths, not only absolute or broken ones.
- **Guess paths for missing textures** — unfound textures get `<subfolder>/<name>`; off keeps their current path. Not available with Strip.
- **Keep .fbx.bak backup** (on by default) — keep it on.

Results and safety

- FBX files are fixed in place. A `<name>.fbx.bak` copy is written next to each file; an existing .bak is kept (so it stays the very first original).
- **Restore backup** (above the path table) restores the selected file from its `.fbx.bak`.
- Changing any option clears the old preview, so preview again before fixing.

- **Rescan** scans again, **Export** saves an HTML / CSV report (e.g. to send an artist the missing textures), **X** clears the list.

4 Blender Packer

Runs Blender 5.0+ headless to analyse and consolidate every dependency of a .blend scene.

How to use

1. **Blender** card: Poseidon finds Blender by itself (`BLENDER_PATH`, PATH, the registry, Program Files\Blender Foundation, Steam). Pick a version in the list (versions before 5.0 are disabled) or **Browse…** to `blender.exe`; ↻ searches again.
2. Drop a `.blend` file on the window (from any tab) or click **Open** in **Pack job**.
3. Choose **Operations** and **Output** (see below).
4. Click **Run pack** (or **Analyse** for analysis only). The progress bar shows each step; **Cancel** stops it. The log shows Blender's output live.
5. The **Report**: Blender version, missing count, dependency count, the **Data-block** / **Path** / **State** table (packed / found / missing, relative / absolute), the zip path and the steps that ran.

Operations

- **Analyse only** — report dependencies without changing anything (disables the writing options).
- **Consolidate** (on by default) — copy external files into `//textures` and remap.
- **Unpack embedded files** — write packed images out to `//textures`.
- **Make paths relative** (on by default) — turn absolute paths into relative ones.
- **Pack everything into the .blend** — embed everything in a single .blend.

Output — where files go

- **Save as a new file** (on by default) — defaults to `<name>_packed\<name>.blend` next to the original; edit it or pick with the folder button.
- Turning **Save as a new file** off **overwrites the original**; Blender keeps the previous version as `.blend1`.
- **Bundle a zip** — packs the .blend with `textures/` and `assets/` into a zip (default `<name>_bundle.zip`).

Tips

- Run **Analyse only** first to see which textures are missing (pink) before packing.
- Only works in the desktop app (it runs a real Blender).

5 Asset Previewer

Quickly inspect 3D meshes and textures (HDR / EXR / PSD included) inside the app, no DCC needed.

How to use

1. Drop meshes, textures or a folder onto the tab, or use the folder / file buttons on the **Assets** card. The 👁 **Preview** buttons in other tabs open files here too.
2. The list on the left: switch list ↔ thumbnails, type in **Filter** to filter by name. The trash button clears the list and cached previews / thumbnails.
3. Select an asset. The bar above shows its path, format, **Auto-reload** (reload when the file — and the model's textures — change on disk; desktop app only) and ↻ **Reload from disk**.

Formats: 3D — FBX, OBJ (+MTL), glTF, GLB. 2D — PNG, JPG, TGA, DDS, WebP, BMP, GIF, TIFF, PSD (composite) and HDR / EXR. .blend files belong to Blender Packer.

3D viewport

`Left drag`	Orbit
`Right drag`	Pan
`Wheel`	Zoom

- View modes: **Shaded**, **Wireframe**, **UV checker**, **Normals**. Toggle **Ground grid**, **Pivot and axes** (X red, Y green, Z blue), **Skeleton** (when there are bones).
- **Materials and textures**: materials and their maps; click a texture to open it in the texture viewer (← returns to the model). Missing textures show in amber with (missing).
- **Lighting and environment**: **Environment** = Studio (built-in), None (lights only), an HDR / EXR from the folder or **Browse for HDR / EXR...**; **Exposure**, **Env strength**, **Key light** (Direction, Height, Intensity), **Light follows the camera**, **Show environment as background**. Double-click a slider to reset it.
- Stats: Triangles, Vertices, Meshes, Materials, UV sets, Bones, Animations, Size (units), File unit; plus missing-texture warnings.
- Animation: Play / Pause, clip picker, timeline, speed 0.25× – 2×.
- **Reset camera**; **Save view as PNG** saves a transparent-background image at 2× resolution, default name `<name>_preview.png`.

Texture viewer

- Drag to pan, wheel to zoom; **Fit to view** and **Actual size (100%)**.
- Channels: RGBA, RGB, A, R, G, B — handy for checking ORM / mask channels.
- **Exposure** (−5 … +5) and **Gamma** (default 2.2), useful for HDR / EXR; double-click to reset.
- Info: **Format**, **Size**, **Data** (float linear or 8-bit sRGB).

6 Pipeline Tools

Six texture and mesh utilities: ORM packer, Converter (left); Duplicates, Unused textures, Texture sets, Meshes (right).

Layout

- The left card switches between **ORM packer** and **Converter**. While Converter is shown, dropped images go to it; otherwise they fill the ORM slots by name.
- The right card analyses a folder: pick a tab, **Choose…** a folder, then click the run button. Dropping a folder on the window runs the selected tool straight away.
- **Export** saves the result as HTML / CSV. Hover a file to reveal **Preview** and **Show in Explorer**.

ORM packer

1. Drop maps named `…_ao`, `…_roughness`, `…_metallic` onto the window (assigned by suffix, including occlusion, rough, gloss, smoothness, metalness…) or click **Pick** on each slot.
2. Tick **Roughness input is a gloss / smoothness map** for gloss maps (ticked automatically when the name says gloss / smooth) — it is inverted.
3. **Output** defaults to `T_<Name>_ORM.png` next to the first map; edit or browse, and choose **PNG** / **TGA**.
4. Click **Pack ORM texture**. The result has Preview and Show in Explorer buttons.

- Channels: **R** = ambient occlusion, **G** = roughness, **B** = metallic. Empty slots use defaults: AO white (1.0), roughness 0.5, metallic black (0.0).
- Smaller maps are resampled to the largest one.
- Note: an existing output file is **overwritten without asking**.

Converter

1. Switch the left card to **Converter**. Drop textures on the window or click **+ File** / **+ Folder** (folders are searched recursively).
2. **Max size** (default 2048 px; **Keep size** = unchanged): only scales down so the longest side fits, keeping the aspect ratio. **Format**: Same as source, PNG, TGA, JPG, DDS (uncompressed + mips).
3. Optionally **Snap to power of two** (e.g. 1000 × 300 → 1024 × 256) and **Flip green channel** (normal maps DirectX / UE ↔ OpenGL / Unity, Blender).
4. **Output folder** (default `converted`, relative to each source's folder, or an absolute path) and **Suffix** (e.g. `_GL`).
5. Click **Dry run** to see the plan (new sizes, skipped files), then **Convert**. **Report** exports the result.

- Existing outputs are skipped unless **Overwrite existing files** is on — including when the output would be the source file itself.
- HDR / EXR sources are skipped (8-bit output would clip them); DDS is not accepted as input. PSD is.

Duplicates

1. Pick the **Duplicates** tab, choose a folder, click **Find duplicates**.

- Files are grouped by size, then hashed with XXH3-128 in parallel; each group is byte-identical files. Shows the group count and the **reclaimable** space if you keep one copy.
- Poseidon deletes nothing — use **Show in Explorer** to deal with them yourself.

Unused textures

1. Pick the **Unused textures** tab, choose a folder, click **Find unused**.

- Every FBX, OBJ / MTL, glTF and GLB under the folder is read and compared with the textures on disk; the result is the textures nothing references, with the reclaimable size.
- Tick **Read .blend files** to include Blender scenes (needs Blender 5.0+, one headless run per file). If some .blend files were not read, textures used only by them appear in the list too — review before deleting.
- Report only; nothing is deleted.

Texture sets

1. Pick the **Texture sets** tab, choose a folder, click **Match sets**: textures are grouped into material sets by suffix (`_BaseColor`, `_N`, `_Roughness`...).
2. The table shows each set with Base color, Normal, ORM, Roughness, Metallic, AO columns (✓ — click to preview; "gloss" when only a gloss map exists) and a **Missing** column.
3. Tick the sets to pack (or **Select all N packable**), choose PNG / TGA, click **Pack N ORM**. The list is rescanned afterwards.

- A set is **ready for ORM** when it has no ORM map yet and at least one of roughness / gloss / metallic / AO. The ORM is written as `T_<Name>_ORM` in the set's folder.
- Textures without a recognisable map suffix are counted separately (hover to see them).

Meshes

1. Drop mesh files on the window while in Pipeline Tools (switches to **Meshes**), or choose a folder and click **Inspect meshes**.
2. Optional **Triangle budget**: files over it are flagged.

- Columns: File (format, mesh count, up axis), Tris, Verts, Mat, UV, Size, Units. Sizes are in file units, before node transforms.
- Typical warnings: no UV set, missing normals, over the triangle budget, unusual unit scale, very large size (check the export scale).
- Files are read directly, no viewport needed — fast for auditing many meshes.

7 Recipes

Chain tools into one repeatable run — in the app or from the command line.

How to use

1. Name the recipe, or pick **Templates...**: Unreal Engine 5 import prep, Unity import prep, Asset audit (read only). The UE5 template is loaded at start.
2. **Add step...** adds a step; the up / down arrows reorder, the trash icon removes. Each step has its own options.
3. **Save** writes the recipe to a `.json` file; **Open** loads a saved one (or drop the .json on the tab).
4. Drop the folder to process on the tab, or click **Choose...**.
5. Keep **Dry run** (on by default) and click **Preview run** — it lists every change without writing anything. When it looks right, untick Dry run and click **Run for real** (the amber button).
6. Review each step's outcome (click a step for details); **Export** saves an HTML / CSV report.

Step types

- **Rename assets** — UE5 preset / Unity preset; **Use Renamer rules** copies the enabled rules from Batch Renamer (shown as Custom); **Include subfolders**. Undo from the Batch Renamer tab.
- **Fix FBX texture paths** — Relative / Strip / Collect (with **Subfolder**), **Rewrite textures that cannot be found**; always keeps `.fbx.bak` backups.
- **Pack ORM textures** — one ORM per texture set with roughness / metallic / AO; PNG / TGA; **Replace existing ORM files**.
- **Convert textures** — max size, format, **Power of two**, **Flip green**, **Overwrite**, **Name contains** (only files whose name contains it), **Output folder**, **Suffix**.
- **Inspect meshes** — report only; **Triangle budget per file**.
- **Find unused textures** (optionally **Also read .blend files**) and **Find duplicates** — report only, nothing is deleted.

Running safely

- Steps run in order. If one fails, the recipe stops ("Stopped after a failed step").
- Always do a **Preview run** first: its result is labelled "Dry run: nothing changed".

Command line

The bottom bar shows the matching command for the current recipe and folder; click the copy icon to copy it. **Save** the recipe first so the command has its file path. Example: `poseidon.exe --recipe ue5-prep.json D:\Assets\Props --dry-run --report report.json`.

- `--dry-run` previews only; `--report <file.json>` writes a report; `--blender <blender.exe>` picks the Blender to use.

- Exit codes: 0 = every step succeeded, 1 = a step failed, 2 = bad arguments or an invalid recipe. In cmd use `start /wait poseidon.exe …` to wait for it.

8 Dashboard

"What is in this delivery?" — asset counts and sizes by type and by package (sub-folder).

How to use

1. Drop an asset folder (local or Google Drive, e.g. `G:\Shared drives\…`) or use **Choose…** / the ☁ menu.
2. Totals: **Assets**, **Total size**, **Packages**, **Meshes**, **Textures**.
3. **By asset type**: Meshes / Textures / Scenes / Sources, with a **By format** breakdown. Hover for details.
4. **By package**: stacked bars per sub-folder. Choose **Top folders** / **2 folder levels** / **3 folder levels**, switch **Chart** / **Table**.
5. **Files** / **Size** switches the metric. Click a package to open it, ↖ **Up one folder** to go back out, **Refresh** to rescan, **Export** for HTML / CSV.

Notes

> - Only file names and sizes are read, so you can survey a whole shared drive without downloading anything.
> - Use it before Asset Matcher to pick the package to work on.

9 Asset Matcher & Kit Assembler

Works out which texture set goes with which mesh / material slot, then assembles a clean kit for UE5 or Unity.

Asset Matcher — how to use

1. Drop a delivery folder (or one package) onto the tab, or use **Choose...** / the ☁ menu.
2. For a Google Drive folder that needs more than 64 MB of meshes read (downloaded), Poseidon asks first: **Read meshes** or **Not now**. Tip: pick a single package to read less. Textures are only listed by name.
3. Each mesh (FBX, OBJ, glTF, GLB) and material slot gets a texture set with a 0–100 confidence: texture paths inside the mesh first, then material and file names, then folders.
4. Filter **All** / **To decide** (Review slots not picked yet and No match slots) / **Review** / **No match**, or type in **Filter mesh, material, set**.
5. Change a set with the slot's dropdown (✎ = picked by hand), or **Accept best for reviews** to take the top candidate for every unpicked Review slot.
6. **Save manifest** writes `asset_manifest.json`; **Export** saves HTML / CSV; **Assemble kit...** opens the Kit Assembler.

Statuses

- **Exact** — the mesh file references this texture set.
- **Likely** — a clear winner by material / mesh name.
- **Review** — a close call or weak evidence: you pick.
- **No match** — no set scored high enough.

Details panel and cache

- Select a slot to see its **Candidates** and reasons; **Use** picks one, **No texture** leaves it empty, **Auto** returns to the matcher's pick. Also lists the set's maps (click to show in Explorer) and the **Texture paths inside the mesh file**.
- With no slot selected it lists **Texture sets no mesh uses** and **Textures without a map suffix**.
- Mesh reads are cached: the next match only reads new or changed files ("N read · M cached"); **clear** forgets the cache. **Refresh** matches again and keeps your picks. **Cancel** while reading — meshes read so far stay cached.

Kit Assembler

1. Click **Assemble kit...** after matching, or **Assemble from manifest...** (when there is no result) / Ctrl+K → **Assemble kit from manifest...** to use a saved `asset_manifest.json`.
2. **Destination (project folder)**, e.g. `<UE project>\Content\Environment`.

3. **Naming**: **Unreal (UE5)** (`SM_Wall_Corner_01.fbx`, `T_Wall_Corner_D.png`, `T_Wall_Corner_ORM.png`), **Unity** (`WallCorner01.fbx`, `WallCorner_BaseColor.png`) or **Keep names**. **Mesh folder** / **Texture folder** default to `Meshes` / `Textures`.
4. The plan (a dry run) appears once a destination is set: **Kind**, **In the kit**, **From**, **Status** (Will write / Kept / …). Check it, click **Assemble N file(s)**, then **Show kit**.

- **Pack ORM maps** (on by default) — roughness / gloss / metallic / AO become one ORM texture; the separate maps are not copied.
- **Relink texture paths in meshes** (on by default) — the FBX, OBJ-MTL and glTF copies point at the copied textures (relative paths).
- **Keep package folders** — `Wall Modular/Meshes/…` instead of one shared `Meshes/` and `Textures/`.
- **Include unconfirmed matches** — unpicked Review slots take the top candidate; off leaves them out ("Left out").
- **Overwrite files already in the destination** — off (the default, and always off when reopened): existing files are kept and skipped.

Safety

- The delivery folder is only read: nothing in it is renamed, overwritten or deleted.
- `kit_manifest.json` is written to the destination as a record. The destination and options are remembered (except Overwrite).
- Suggested flow: Dashboard → Asset Matcher → Kit Assembler → UE5 / Unity project.